Engineering Chemistry · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry

DRAFT — UE26CY151A · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry · Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
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Low ASR confidence	Transcript segment requires audio verification.
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M1

Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential

1 Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential

Study focus: Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000146 · EV0000147

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will start with unit two.

Key details

- First, we will look at module three on phase equilibrium.
- The module content goes like this.
- We will understand what is phase equilibria.
- We will derive Gibbs phase rule.
- Using the Gibbs phase rule, we will construct the phase diagram of a one component system and then go on to apply the Gibbs phase rule to a two component system and construct the phase diagram.
- And then we'll study the iron carbon phase diagram.
- We will introduce a new term called chemical potential and then we look up certain definitions in phase equilibria that is about the phase component and degree of freedom.
- Free energy in phase equilibrium.
- What exactly do we mean by phase equilibrium?
- We have different phases of matter, you're aware, of solid, liquid, and gas.
- And we find that in certain conditions the solid phase is more stable.
- In other conditions, liquid phase is more stable, and in some other conditions, gaseous phase becomes more stable.
- We know that below zero degrees we find solid phase ice is stable.
- At from 0 to 100, we find it's the liquid phase, and then beyond 100, it is the water vapor.

- What is the driving force?
- Why is it that below zero degrees we find only the solid phase and not the liquid phase?
- Whichever phase has the lowest free energy, that phase is going to be present.
- We know that all systems try to reduce their free energy and.
- In this case, we find that solid is having lower free energy than liquid at zero degrees.
- When you cross zero and up to 100 degrees, we find that the liquid phase has minimum free energy, and similarly, gaseous phase will have less lower free energy when it comes to beyond 100 degrees.
- So every system tries to get minimum free energy, and therefore the phase which has minimum free energy will be most stable phase.
- That is, solid changes into liquid.
- Ice is getting transformed into the liquid phase.
- beyond the melting point.
- When you heat the system beyond the melting point, it the solid changes into liquid.
- Why would such a transformation happen?
- Between 0 to 100, it is a liquid phase which is more stable.
- And what would that stable phase mean?
- Just now I said it will have a lower free energy.
- which has a higher free energy to one which has lower free energy.
- Reduce free energy of the system.
- That means Gibbs free energy plays a very important role in deciding what phase transformations will happen under a certain set of conditions.
- But Gibbs energy is an extensive property.
- What is chemical potential?
- It's basically gives free energy divided by the number of modes.
- The moment you do that, you have made it into an intensive property, and now you can compare systems, phases, etc, with the help of this new term which is known as.
- So just saying gives free energy divided by the number of moles that is molar Gibbs free energy would not be sufficient because you have so many components, we say chemical potential, we define it as partial molar Gibbs free energy.
- It is not just Gibbs free energy, it is partial molar Gibbs free energy.

Example or application from the lecture

- The most common example we can take is that of water.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

Core explanation · 05:00–09:59

How would I mathematically give an expression for this partial molar gives free energy.

Key details

- And remember it is denoted by a symbol mu.
- So for this component chemical potential is defined as del G by del Ni.

- NI is the number of moles of that particular component in the system.
- And since a partial derivative, so pressure, temperature, and number of moles of other components are taken as constant.
- Whenever we say phases are in equilibrium, now instead of Gibbs free energy, we'll refer to chemical potential.
- So, what would be the condition for phases to be in equilibrium?
- The chemical potential of a particular component should be the same in all the phases which are present in equilibrium.
- So that means at the triple point of water where you have solid, liquid, and the vapor in equilibrium,.
- What is the condition for phase equilibrium?
- The chemical potential of the solid form is equal to the chemical potential of the liquid form, which is equal to the chemical potential of the vapor form.
- what would happen in that case, then at some point in some phase the chemical potential is more as compared to the chemical potential in another phase.
- So what would happen because of that?
- There would be a transformation, phase transformation from higher chemical potential to lower chemical potential.
- If it is a heterogeneous system, of course, when we say different phases are present, it is a heterogeneous system.
- There can be no transfer of energy or mass from one phase to another.
- Only when there is no transfer of mass or energy, we say the system is at equilibrium, or.
- Another way of looking at it is when a system is in equilibrium, the various phases must have the same temperature and pressure, and the composition must remain constant all along.
- These are the different ways by which we can understand phase equilibrium.
- A phase is defined as a homogeneous and physically distinct part of a system.
- And there will always be a boundary between phases.
- Whenever you see a boundary, when one phase from the other, there will be a boundary.
- And it is denoted by P, and P is a number.
- Whenever we say phase, we are referring to the number of phases in a particular system.
- Now, if we have a liquid system, the number of layers, we know that when liquids are immissible, the layers going to be seen in that system, and therefore the number of layers will decide the number of phases.
- Provided they are invisible.
- When we have a solid state, then each state, each of these solid will constitute a separate.
- Phase we have to count each solid as a separate phase.
- All solids are not taken as one phase because how did we define phase?
- There is a boundary.
- They are mechanically separable.
- Therefore, each solid is going to constitute one phase.
- And not only that, each polymorphic form or allotropic form.

Example or application from the lecture

- Three phases, then in that case we find that suppose it is solid, liquid, and vapor, then we find what is the requirement for a system to be at equilibrium, that the chemical potential the chemical potential of a component is equal in all the phases which are present in equilibrium.
- First, let us take up phase.
- If they are completely miscible, for example, alcohol and water, then P is going to be equal to one.

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EV0000150 · EV0000151 · EV0000152 · +1 source units in pack.json

Core explanation · 10:00–14:59

Is also going to constitute a separate phase.

Key details

- What do we mean?
- Carbon exists as graphite as well as diamond.
- Both of them cannot be treated as one phase, or we cannot say p is equal to one here.
- Because both of them have boundaries, they can be easily separated.
- And if there is a solid solution as in isomorphic alloys, then P is taken as one.
- Here we see P is equal to C three.
- Because it may be the same system.
- It may be.
- It may be water system.
- Remember that two solids will contribute to two different phases and one gas.
- So we have two solid phases, calcium carbonate, calcium oxide, and P is equal to three now because there's also carbon dioxide gas.
- In this case, again, P is equal to 2.
- Because solid is one, but gases are completely miscible, therefore ammonia and Cl together will constitute one phase.
- A component is defined as the smallest number.
- We need to pay attention to these.
- parts which have been underlined, smallest number of independently varying chemical constituents using which the composition of each and every phase.
- These two are to be marked very carefully.
- Smallest number and each and every phase of the system we should be able to describe with the help of these components.
- This is a number and is denoted.
- C will be the number of components.
- And when there is no reaction taking place, obviously as many number of constituents will be equal to the number of components.
- And since a number, one important.
- Fact which we have to remember is that when we are expressing in terms of constituent zero and negative values are also allowed.

- Water is a one component system where C is equal to one.
- Even if say we have two phases in equilibrium, that is at the melting point, the composition of both the phases will be the same.
- If since it's a single constituent, there is both of them will have the same composition.
- Both of them will be the same.
- We may have what are the phases here?
- What are the number of phases?
- We have two phases here.
- Solid water as well as liquid water, but both can be expressed in terms of a single constituent.
- That is a salt hydration.
- Salt hydrate system.
- Salt hydrate meaning what?
- It has a water of hydration.
- Because NA2SO410H2O can be expressed as the sum of NA2SO4 plus 10H2O.
- Only Na2SO4, if, that is not sufficient for me to describe this phase.
- Another phase of the same salt hydrate could be Na2SO47H2.
- We will write it as Na2SO4 plus 7H2.

Example or application from the lecture

- For example, we have carbon.
- Like this is given by this equation, what is the number of phases?
- Now let's come to the next definition, and that is of a component.

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EV0000154 · EV0000155 · EV0000156 · +1 source units in pack.json

Core explanation · 15:00–19:59

If we take the anhydrous salt, then what will happen?

Key details

- NE2SO4 is NE2SO4 plus 0H2O.
- As I mentioned, in the case of components, 0 can also be used.
- We are calling it a two-component system.
- One component is enough to describe this particular phase, but we define component as what?
- The total number of phases, the minimum number, the smallest number which is described which is used to describe all the phases.
- You should be able to describe all the phases using that minimum number.
- Here, though it is one, minimum you need is two to describe.
- All the phases.
- Evaporate the water completely and collect it, then you will have only the water phase.
- Meaning how does or how do all these phases come about here?

- Because you are changing the conditions, you can have different forms of the same salt hydrate.
- And how do we describe these phases?
- In order to describe the phases, now when you come to water, you will say zero of any two so four and water.
- So that means you need at least two components to describe this particular system.
- Again here we say C is equal to 2.
- Because though we have three different phases or three different constituents, there's an equation which is binding them together.
- We need not use all the three constituents.
- Two components are sufficient to describe all the phases.
- What are the phases?
- We have calcium carbonate solid, calcium oxide solid, and CO2 gas.
- Then the first phase CACO3 solid will be described as CAO plus CO2.
- CAO itself will be described as CAO plus 0CO2 and CO2 can be described as CAO plus CO2.
- How will I describe the phases now?
- We have CACO3 and CO2.
- But if I have to describe CAO now, I don't have CAO with me.
- How will I use these two?
- I can say CACO3 minus of CO2.
- In this case two components are sufficient for me to describe all the three phases.
- Here, though the reaction looks similar to the calcium carbonate reaction, C becomes equal to 1.
- What are the phases?
- Here we have only two phases.
- Remember, the one is a solid phase of ammonium chloride, and the other is the gaseous phase where both ammonia and HCl are present in the molar ratio 1 is to 1.
- Mixture of NH3 plus H Cl, but the num the phase is only one because gases are completely miscible.
- In this case, the solid phase is ammonium chloride.
- But when we come to the gaseous phase, we have ammonia and HCl in equal quantities.
- So therefore, ammonia.
- A plus HCl can also be represented as ammonium chloride.
- That's so in this case one component is sufficient for me to describe okay.
- I could use two components also.
- What ammonia and HCl.
- If I want to use ammonia and HCl, it will become ammonia plus HCl in this case, and the gaseous form will also be ammonia plus HCl.
- But the by definition component is the smallest number of constituents which are required.

Example or application from the lecture

- Let us see.
- Let us see if we can describe all the three phases using any two constituents.

- Let us take up another case where we have thermal decomposition of ammonium chloride in a closed system.

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EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 20:00–24:59

The composition of both these phases can be described with C is equal to one.

Key details

- But this is if it was a closed system.
- You have added certain extra amount of HCl or ammonia, so that has to be accounted for.
- Therefore, C becomes equal to 2 in this case.
- Just ammonium chloride is enough for the solid phase, but not enough for the gaseo phase because we added an extra amount of HCl or ammonia.
- The degrees of freedom or variance of a system is defined as the minimum number of intensive variables, such as temperature, pressure, or concentration, that must be ascertained in order to describe the system.
- In other words, we can also say what is my degree of freedom to change these variables independently without altering the number of phases, or rather without altering the equilibrium in the system.
- At a particular time, there is only one phase, so solid, liquid, or the gas.
- the without affecting the system, I can vary the temperature or pressure.
- In fact, solid, liquid, or gas form of water can be present at different temperature and pressure.
- Therefore, the degree of freedom or the variance is F that is equal to 2.
- Two phases in equilibrium where now temperature or pressure can be.
- They read or can be specified.
- What does it what do we mean?
- The degree of freedom now becomes one.
- When does this happen?
- This happens at zero degrees centigrade, but the moment I say zero degrees centigrade, the pressure becomes fixed to one atmosphere.
- If I change the pressure, then the melting point is also going to change.
- That zero degrees will not remain zero degrees.
- Though we have solid water and liquid water in equilibrium, now I can either change the temperature or I can change the pressure.
- Therefore, when two phases are in equilibrium, f becomes equal to one.
- Here, none of the variables can be changed, meaning that neither the temperature nor the pressure can be changed, and therefore we say f is equal to.
- That would be a triple point of water.
- Solid water is in equilibrium with liquid water, which is in equilibrium with water vapor.
- All the three phases are present together, they coexist in equilibrium, and now.
- Neither the temperature nor the pressure can be changed because those are fixed quantities.

- And what is that value?
- 0.0098 degrees centigrade and 4.6 millimeter of mercury for water.
- In a system, F becomes equal to 2.
- You can change temperature and pressure independent of each other.
- When you come to two phases in equilibrium, in that case, you can either change the temperature or you can change the pressure so f is equal to one and when you have three phases in equilibrium then f becomes equal to zero.
- You can neither change the temperature nor you can change the pressure.
- Remember they are numbers.
- So now in the next class we are going to look at.
- Gibbs phase rule, which is going to combine all these three components, phase and degree of freedom, and give you a very nice elegant thermodynamic equation.

Example or application from the lecture

- Let us look at how to understand this.
- Let us take up the water system.
- Suppose we have two phases in equilibrium.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000162 · EV0000163 · EV0000164 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__a7da1e88f96b163b.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__a7da1e88f96b163b.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__a7da1e88f96b163b.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Phase, Component, Degrees of freedom, Statement of Gibb's phase rule, chemical potential transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/008_Phase_Component_Degrees_of_freedom_Statement_of_Gibb_s_phase_rule_chemical_potential__a7da1e88f96b163b.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00030	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00031	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00032	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00033	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

C Revision and retrieval practice